

PUNEETH KALYAN

IIIT Bangalore

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Education

Mahesh Pre-University College

July 2019 – June 2021

AIR 6984 in JEE Mains 2021, KCET '21 Rank 646, State Examination- 90%

International Institute of Information Technology Bangalore (IIIT-B)

November 2021 – July 2027

Integrated Dual Degree (BTech + MTech) in Electronics and Communication Engineering

CGPA: 2.74/4.00

Projects

RAG Q/A Assistant – Inter-domain Ecosystem | Python, LangChain, FAISS, BGE-M3, Ollama Aug '25- Apr '26

- @Web Science Lab, IIITB - Built a domain-specific, end-to-end, offline RAG pipeline over a vast multi-domain knowledge base surrounding the Karnataka Sandalwood ecosystem, implementing multi-signal re-ranking across semantic similarity, keyword overlap, and source quality scores, complemented by a confidence threshold gating layer and inline source attribution.

Information Convergence Platform | Flask, MySQL, XML

March 2025

- Developed a unified querying engine to integrate heterogeneous data sources including MySQL, XML and XLSX using a declarative JSON-based query format.
- Designed and implemented a universal schema representation to abstract and map local schemas across disparate data types, enabling seamless data retrieval.
- Built a Flask based web interface to allow users to upload schemas, run complex join queries across formats, and visualize results as tables or JSON.

Codeforces Problem Recommender | Python, PyTorch, Javascript

May 2024

- A personalized problem recommender that tailors recommendations to users based on their individual profiles.
- Leveraged the Codeforces API to curate a rich dataset of user profiles and problem-solving histories and implemented a Seq2Seq LSTM model in PyTorch for next problem prediction.
- Achieved a 14x performance boost in NDCG@5 score compared to a popular recommendations baseline model.

Distributed RDF Store System | Java, MongoDB, PostgreSQL, Neo4j

April 2024

- Built a distributed Resource Description Framework (RDF) triple store system in Java using OOP principles.
- Utilized PostgreSQL, MongoDB, and Neo4j for backend data storage and management of triples, added indexing to improve query speed by 1000 times on 13 million rows from the YAGO dataset.
- Developed methods for querying, updating, merging data, conflict resolution, and sharding log files to ensure scalability and data consistency.

Technical Skills

Languages: Python, Java, C++, C, SQL, HTML, CSS, JavaScript, XML

Technologies/Frameworks: LangChain, FAISS, Ollama, Spring Boot, PostgreSQL, MongoDB, Neo4j, PyTorch, numpy, pandas, Linux, GitHub

Relevant Coursework

Data Structures, Algorithms, Object Oriented Programming, Computer Networks, Computer Architecture, Natural Language Processing, Operating Systems, Digital Electronics, Analog Circuits, Technology Regulation & Policy.

Extracurriculars

Infin8- IIIT Bangalore

- * Organised an inter-college sports competition, managing a team of 8 people to ensure a seamless event.

Chess Club- IIIT Bangalore

- * Actively participated in the promotion and functioning of the club and conducted regular tournaments.

Social Outreach Volunteer- IIIT Bangalore

- * Played a key role in organizing the distribution of household essentials to an underprivileged neighborhood by leading logistics and communication efforts. Leveraged fluency in four languages to coordinate with neighborhood residents and student volunteers.